

Risk and Logistics A1

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Aggregation

For our Multi-Echelon Capacitated Warehouse Location Problem (MECWLP) model, we have to assign 440 Customers to one of 440 Warehouse candidate locations in each of the 10 time periods. The combinatorial nature of this problem means that the number of binary assignment variables grows in proportion to $|J| * |C| * |T|$. This means that the binary assignment variable $Z_{i,j,t}$, where $i, j \in \{0, 1, \dots, 440\}$ and $t \in \{1, 2, \dots, 10\}$ has $440 * 440 * 10 = 1,936,000$ binary variables.

As the Branch-and-Bound method used by MIP solvers explores combinations of these binaries, memory and time requirements can grow exponentially in worst case scenarios. As a result, in order to ensure that our model can run effectively, we have chosen to aggregate our customer market in order to answer the first question.

We chose to aggregate customers rather than candidate warehouse locations because customers do not correspond to strategic investment decisions. Instead, they represent spatially distributed demand. By aggregating customers, we reduce the dimensionality of the assignment component of the model while preserving:

- Total demand volume,
- The spatial distribution of demand,
- The economic structure of warehouse investment decisions

In contrast, aggregating candidate warehouse locations would directly distort the strategic decision space. Each candidate location has its own fixed setup cost, operating cost, and capacity. Combining or averaging these locations could eliminate feasible alternatives or alter the investment structure of the problem. This would fundamentally change the optimisation landscape and potentially bias the warehouse location decisions.

To aggregate customers meaningfully, we required a clustering method that preserves the economic structure of transportation costs rather than relying purely on geographic proximity.

We did this by constructing a candidate-customer cost matrix, X , such that:

$$X(i, j) = \text{cost of sending 1 kg from candidate } j \text{ to customer } i$$

This matrix captures the true economic transport structure of the network where each row represents a cost profile for a customer.

We then centered each customer's cost profile by subtracting its mean:

$$cX = X - \text{mean}(X)$$

By doing this we ensured clustering was done based on relative cost differences rather than absolute cost levels. Otherwise customers in remote areas would appear artificially similar simply because all candidate warehouses are uniformly expensive for them. Centering enabled us to capture which warehouses were relatively cheaper or more expensive for each customer.

Using this, we constructed a customer-customer distance matrix using the squared Euclidean distance:

$$D_{ij}^2 = \|x_i - x_j\|^2.$$

Computationally, this was implemented via:

$$D^2 = \text{sq} + \text{sq}^T - 2G,$$

where:

$$G = XX^T, \quad \text{sq} = \sum_k X_{ik}^2.$$

The resulting matrix D measures how similar two customers are in terms of their warehouse cost profiles.

- A small $D(i, j)$ indicates similar cost structures.
- A large $D(i, j)$ indicates substantially different cost structures.

With this, customers could now be clustered based on economic similarity rather than purely geographic proximity.

We used K-medoids clustering in our model due to its robustness to extreme cost profiles, its compatibility with operating directly on pairwise (cost) distance matrices and the fact that each chosen mediod corresponds to a real customer location, which helps with interpretability and realism.

We employed K-medoids++ initialisation to improve clustering quality. The first medoid is selected randomly. Subsequent medoids are chosen with probability proportional to their distance from the nearest existing medoid. This reduces the likelihood of selecting multiple medoids from the same region while leaving remote regions underrepresented.

Whilst the aim is to reduce the number of customer points in order to reduce the computation complexity of the MIP model, we must balance this with ensuring that we do not lose valuable data and that those customers who are aggregated together do indeed have similar cost profiles. Excessive aggregation runs the risk of distorting the demand structure and cost structure of the customers. To determine a suitable number of clusters we evaluated three metrics:

- **Demand-Weighted Distortion:** This measures the demand weighted clustering error. We use weighting because misrepresenting a high-demand customer is more costly to our model than misrepresenting a low-demand customer. A small value indicates that aggregation introduces minimal system-wide distortion.
- **Maximum Within-Cluster Variation:** This measures the worst-case deviation within any cluster. Large values in this metric indicate an ineffective aggregation of customers and a potential loss of important information.
- **Stability:** This measures how susceptible the clustering is to initialisation (ie. the first choice of mediod). A high score here indicates high susceptibility, whilst a low score suggests that the clustering structures are robust.

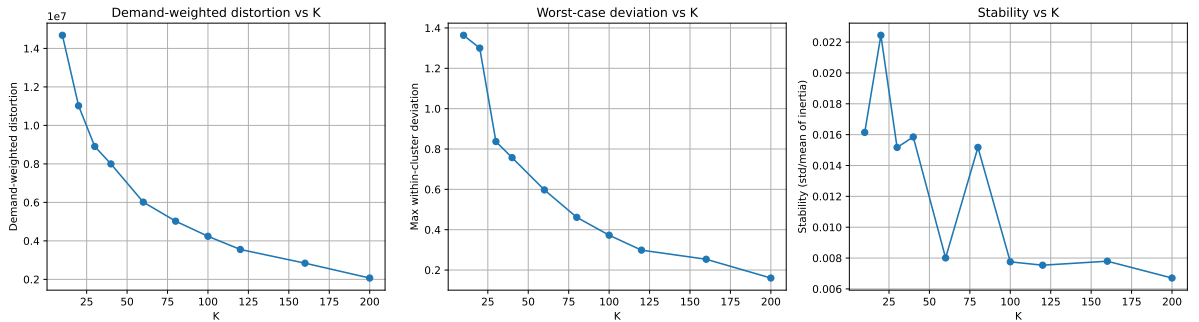


Figure 1: Broad Search - diagnostic metrics vs number of clusters K .

To determine an appropriate number of clusters K , we plotted each diagnostic metric against K and searched for 'elbows'. These are points of diminishing returns where further increases in K produce only marginal improvements in clustering quality. To identify these elbows objectively, we applied a geometric knee-detection method.

After normalising both the number of clusters and the corresponding metric values, we computed the perpendicular distance of each point to the straight line connecting the smallest and largest tested

values of K . The value of K associated with the maximum perpendicular distance was selected as the point of maximum curvature. This represents the best trade-off between aggregation fidelity and model complexity.

Since elbow locations can be sensitive to the range of K considered, we first conducted a broad search over a wide range of cluster values. This preliminary analysis suggested that the elbow lay approximately within the interval $K \in [40, 100]$. We then performed a more detailed search within this narrower range.

The demand-weighted distortion metric indicated an elbow around $K = 75$, while the maximum within-cluster deviation exhibited a clear elbow at $K = 70$. Considering both metrics together, and noting that $K = 75$ lies immediately beyond the elbow region while providing slightly improved distortion performance, we selected $K = 75$ as a practical compromise.

Although the stability metric did not display a pronounced elbow, its values remained consistently low across the tested range. At $K = 75$, the stability measure was just above 1%, indicating that the clustering solution is robust to initialisation and therefore reliable.

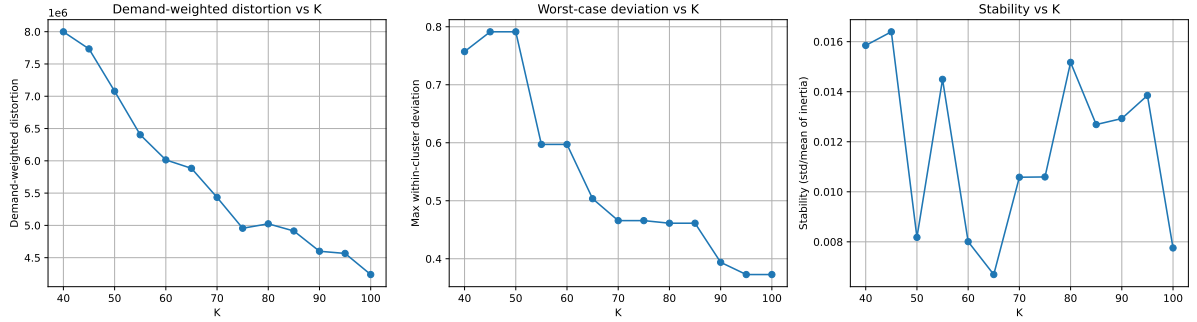


Figure 2: Detailed Search - diagnostic metrics vs number of clusters K .

Having selected $K = 75$, we proceeded to aggregate the customer data accordingly.

Aggregated Demand. For the first problem, the aggregated demand of each cluster, product, and period was calculated by summing the demand of all customers assigned to that cluster. Formally, for cluster c , product p , and period t , aggregated demand is given by

$$D_{c,p,t} = \sum_{i \in \mathcal{C}(c)} D_{i,p,t}$$

where $\mathcal{C}(c)$ is the set of customers belonging to cluster c , and D_{ipt} is the demand of customer i for product p in period t . This preserved total system demand exactly.

For the second problem we aggregated demand similarly, but also by scenario ω too:

$$D_{c,p,t,\omega} = \sum_{i \in \mathcal{C}(c)} D_{i,p,t,\omega}$$

Aggregated Transportation Costs. To compute transportation costs from candidate warehouse j to aggregated cluster c , we constructed a demand-weighted average of individual customer costs. Specifically, the aggregated cost was calculated as

$$\tilde{c}_{j,c} = \frac{\sum_{i \in \mathcal{C}(c)} w_i c_{j,i}}{\sum_{i \in \mathcal{C}(c)} w_i},$$

where:

- $c_{j,i}$ is the transportation cost from candidate warehouse j to customer i ,
- w_i is the total demand weight of customer i (across products and periods),
- $\mathcal{C}(c)$ is the set of customers in cluster c .

This method means that customers who have a larger overall demand exert a greater influence on the aggregated transportation cost. Therefore, the aggregated cost accurately reflects the economic importance of high-demand customers within each cluster.

Question 2

The formulation for our MECWLP with multiple time periods

Our customers are aggregated in 75 clusters as explained above. For our warehouses, we consider all 440 candidate locations.

Sets:

- $T := \{1, \dots, 10\}$ - Time periods
- $I := \{1, \dots, 75\}$ - Customers (aggregated)
- $J := \{1, \dots, 440\}$ - Locations
- $S := \{1, \dots, 53\}$ - Suppliers
- $P := \{1, \dots, 4\}$ - Products

Data:

- f_j - Fixed cost of setting up warehouse $j \in J$
- g_j - Annual operating cost of warehouse $j \in J$
- b_{ip}^t - Demand of customer $i \in I$ for product $p \in P$ in time period $t \in T$
- c_{ij} - Cost of serving customer $i \in I$ from warehouse $j \in J$
- d_{sj} - Cost of supplying warehouse $j \in J$ from supplier $s \in S$
- k_j - Capacity of warehouse $j \in J$
- L_s - Capacity of supplier $s \in S$

Decision variables:

- $x_{ij}^t \in \{0, 1\}$ - Allocates customer $i \in I$ to warehouse $j \in J$ in time period $t \in T$
- $y_j^t \in \{0, 1\}$ - Decision to build warehouse $j \in J$ in time period $t \in T$
- $z_j^t \in \{0, 1\}$ - Decision to operate warehouse $j \in J$ in time period $t \in T$
- $w_{jsp}^t \in \mathbb{Z}^+$ - Number of products $p \in P$ sent by supplier $s \in S$ to warehouse $j \in J$ at time period $t \in T$

Objective function:

$$\min \sum_{j,t} (f_j y_j^t + g_j z_j^t) + \sum_{i,j,t,p} b_{ip}^t x_{ij}^t c_{ij} + \sum_{s,j,t} w_{sj}^t d_{sj}$$

Our objective function aims to minimise the total cost of our procedure. The first summation represents the setup and operating costs of any warehouses we build. The second summation represents the costs involved in transporting goods from warehouses to customers. The third summation represents the costs involved in transporting goods from suppliers to warehouses.

Constraints:

$$\sum_j x_{ij}^t = 1 \quad \forall i \in I, \forall t \in T$$

Each customer will be assigned to exactly 1 warehouse per time period

$$x_{ij}^t \leq z_j^t \quad \forall i \in I, \forall j \in J, \forall t \in T$$

Customers can only be assigned to operational warehouses

$$z_j^t = \sum_{u \in T, u \leq t} y_j^u \quad \forall j \in J, \forall t \in T$$

Warehouses only operate in time period t if built in time period t or before

$$\sum_{p,j} w_{jsp}^t \leq L_s \quad \forall s \in S, \forall t \in T$$

Goods supplied by each supplier does not exceed that suppliers capacity

$$\sum_{p,i} b_{ip}^t x_{ip}^t \leq k_j z_j^t \quad \forall j \in J, \forall t \in T$$

Total demand serviced by each warehouse does not exceed that warehouses capacity

$$\sum_w w_{jsp}^t = \sum_i b_{ip}^t x_{ij}^t \quad \forall j \in J, \forall t \in T, \forall p \in P$$

Our total supply must equal our total demand

$$x_{ij}^t, y_j^t, z_j^t \in \{0, 1\} \quad \forall i \in I, \forall j \in J, \forall t \in T$$

Domain constraint

$$w_{jsp}^t \in \mathbb{Z}^+ \quad \forall i \in I, \forall j \in J, \forall t \in T, \forall s \in S$$

Domain constraint

Results:

We found an optimal solution had an optimal value of £31,642,786.73 ($\approx$ £31.6M), split as such:

Project Area	Total Cost
Construction	£5,370,000
Operational	£4,205,200
Transport	£22,067,586.73

Our optimal solution builds a total of 6 warehouses in the following schedule:

Time period	Candidate warehouse built
1	19 and 315
3	42 and 183
5	48
7	54

In all other time periods, we construct no extra warehouses. There is a fairly even split across time periods as we balance building enough warehouses to fulfil early demand whilst keeping construction and operating costs as low as possible (i.e. building fewer warehouses). Our total runtime to find an optimal solution was 5 minutes.

As we can see, the majority of our costs come from transportation. Our formulation has largely decided that building extra warehouses is not worth the reduction in transportation costs.

We decided to have each customer get all their products from exactly one warehouse. We did this for simplicity, and well as traceability; it becomes easier to diagnose a customer's problem if we know exactly where all their products came from. We don't impose this same constraint on suppliers, however. Firstly, this is because each supplier only supplies 1 product, and so warehouses will need a minimum of 4 suppliers to stock all products. The second, is that in a real world scenario, customers are going to have preferences over products, and the brand of product 1 that one customer likes might be different to that of their neighbour. Therefore, it made sense to have warehouses supplied by a range of suppliers. That being said, our formulation will obviously prioritize those with lower transportation costs.

Question 3

Formulation for the MECWLP with multiple time periods and uncertain demands

We decided to use $N_\Omega = 10$ scenarios, as opposed to the full 100 scenarios available, in order to keep the model computationally solvable. We found this was the maximum number of scenarios we could

consider before our computers ran out of RAM (8GB). Perhaps this a sign that our formulation could be improved in areas to focus on computational efficiency.

We once again used $K = 75$ clusters to aggregate our customers, but left warehouses disaggregated as to not let the model stray too far from reality. We felt that aggregating the warehouses constrained the candidate locations to a set which didn't properly represent the problem.

Sets:

- $T := \{1, \dots, 10\}$ - Time periods
- $I := \{1, \dots, 75\}$ - Customers (aggregated)
- $J := \{1, \dots, 440\}$ - Locations
- $S := \{1, \dots, 53\}$ - Suppliers
- $P := \{1, \dots, 4\}$ - Products
- $\Omega := \{1, \dots, 10\}$ - Scenarios

Data:

- f_j - Fixed cost of setting up warehouse $j \in J$
- g_j - Annual operating cost of warehouse $j \in J$
- $b_{ip}^{t\omega}$ - Demand of customer $i \in I$ for product $p \in P$ in time period $t \in T$ and scenario $\omega \in \Omega$
- c_{ij} - Cost of serving customer $i \in I$ from warehouse $j \in J$
- d_{sj} - Cost of supplying warehouse $j \in J$ from supplier $s \in S$
- k_j - Capacity of warehouse $j \in J$
- L_s - Capacity of supplier $s \in S$

Decision variables:

- $x_{ij}^{t\omega} \in \{0, 1\}$ - Allocates customer $i \in I$ to warehouse $j \in J$ in time period $t \in T$ and scenario $\omega \in \Omega$
- $y_j^t \in \{0, 1\}$ - Decision to build warehouse $j \in J$ in time period $t \in T$
- $z_j^t \in \{0, 1\}$ - Decision to operate warehouse $j \in J$ in time period $t \in T$
- $w_{jsp}^{t\omega} \in \mathbb{Z}^+$ - Number of products $p \in P$ sent by supplier $s \in S$ to warehouse $j \in J$ at time period $t \in T$ and scenario $\omega \in \Omega$

Objective function:

$$\min \sum_{j,t} (f_j y_j^t + g_j z_j^t) + \sum_{\omega} \frac{1}{10} \left[\sum_{i,j,t,p} b_{ip}^{t\omega} x_{ij}^t c_{ij} + \sum_{s,j,t} w_{sj}^{t\omega} d_{sj} \right]$$

As before, our objective function aims to minimise the total cost of our procedure. The first summation represents the setup and operating costs of any warehouses we build. The second summation represents the transport costs between suppliers, warehouses and customers. This summation is averaged over all possible scenarios. Since we are assuming that each of our scenarios are equally likely, we have a constant weighting term, $\frac{1}{N_{\Omega}=10}$.

Constraints:

$$\sum_j x_{ij}^{t\omega} = 1 \quad \forall i \in I, \forall t \in T, \forall \omega \in \Omega$$

Each customer is assigned to exactly 1 warehouse per time period

$$x_{ij}^{t\omega} \leq z_j^t \quad \forall i \in I, \forall j \in J, \forall t \in T, \forall \omega \in \Omega$$

Customers can only be assigned to operational warehouses

$$z_j^t = \sum_{u \in T, u \leq t} y_j^u \quad \forall j \in J, \forall t \in T$$

Warehouses only operate in time period t if built in time period t or before

$$\sum_{p,j} w_{jsp}^{t\omega} \leq L_s \quad \forall s \in S, \forall t \in T, \forall \omega \in \Omega$$

Goods supplied per supplier does not exceed that suppliers capacity

$$\sum_{p,i} b_{ip}^{t\omega} x_{ip}^{t\omega} \leq k_j z_j^t \quad \forall j \in J, \forall t \in T, \forall \omega \in \Omega$$

Total demand serviced by each warehouse does not exceed that warehouses capacity

$$\sum_w w_{jsp}^{t\omega} = \sum_i b_{ip}^{t\omega} x_{ij}^{t\omega} \quad \forall j \in J, \forall t \in T, \forall p \in P, \forall \omega \in \Omega$$

Our total supply must equal our total demand

$$x_{ij}^{t\omega}, y_j^t, z_j^t \in \{0, 1\} \quad \forall i \in I, \forall j \in J, \forall t \in T$$

Domain constraint

$$w_{jsp}^{t\omega} \in \mathbb{Z}^+ \quad \forall i \in I, \forall j \in J, \forall t \in T, \forall s \in S$$

Domain constraint

Results:

Our final solution had an objective value of £33,578,252.92 ($\approx$ £33.6M), split as such:

Project Area	Total Cost
Construction	£8,082,000
Operation	£5,311,800
Transport	£20,184,452.92

Our final solution involved building a total of 10 warehouses over the following schedule:

Time period	Candidate warehouse built
1	42 and 315
2	-
3	183 and 411
4	48
5	-
6	54
7	19 and 339
8	47
9	351
10	-

Our MIP gap was 0.73%, so whilst our solution has not been formally proven optimal, it is extremely close. With such a small MIP gap, we can be confident that this is a good solution to our MECWLP problem. Our total run time was 12 hours. Considering the fact these are important decisions that only need to be made once, at the beginning of the project planning phase, this is not an unreasonable time. It is better to wait longer for a better solution to these sort of setup decisions than to make our model worse in pursuit of a quicker run time.

In our simpler model without uncertain demand, we built a total 6 warehouses. It is interesting to note that we build the same 6 warehouses again this time, but now with 4 extra locations. We don't, however, build them in exactly the same order. It is no surprise that we build the same warehouses; those warehouses have proved to be optimal for this set of customers and so it follows that they would continue

to be useful, even with stochastically in the demand. Since we have ensured that every subproblem is feasible (i.e. we can meet demand in every scenario) We are forced to build more warehouses in this model, since we must account for the scenarios in which demand is abnormally high.

In the stochastic model, we out-spend our simpler model for construction and operational costs by approximately £2.6 million and £1.1 million respectively. This is to be expected, since we're building more warehouses, and more generally we are accounting for situations in which we need to supply, store and deliver more products. The transportation costs in the stochastic model, however, are approximately £2 million less than our simpler model. This also makes sense as, since we are building more warehouses, we are going to average shorter journeys between any pair of customers, warehouses or suppliers. The balances out to our stochastic model having an objective value approximately £2 million higher than our simpler model.

Question 3c

Making our model less conservative

One approach we could take to make our formulation less conservative is to penalize for not meeting all of our demand. This is a sensible choice because the main issue of our previous model were that high-demand scenarios were forcing our formulation to build lots of warehouses in order to always meet demand. This was overly conservative, accounting for a very small group of high-demand scenarios and forcing the whole system to compensate. To implement this using our previous formulation, we would first remove the constraint:

$$\sum_w w_{jsp}^{t\omega} = \sum_i b_{ip}^{t\omega} x_{ij}^{t\omega} \quad \forall j \in J, \forall t \in T, \forall p \in P, \forall \omega \in \Omega \quad (1)$$

This constraint ensures that the amount of goods we supply in each time period is equal to the total demand we need to service in that time period. $\sum_w w_{jsp}^{t\omega}$ represents our total supply and $\sum_i b_{ip}^{t\omega} x_{ij}^{t\omega}$ represents our total demand. Without this constraint, our supply variables, $w_{jsp}^{t\omega}$, have no constraints put on them (other than domain constraints) and so the formulation will set all these variables to zero in order to "ignore" the transport costs between suppliers and warehouses. To account for this, we must amend our objective function to:

$$\min \sum_{j,t} (f_j y_j^t + g_j z_j^t) + \sum_{\omega} \frac{1}{10} \left[\sum_{i,j,t,p} b_{ip}^{t\omega} x_{ij}^{t\omega} c_{ij} + \sum_{s,j,t} w_{sj}^{t\omega} d_{sj} + \sum_{j,t,p,\omega} \lambda \left(\sum_i b_{ip}^{t\omega} x_{ij}^{t\omega} - \sum_s w_{jsp}^{t\omega} \right) \right]$$

The first three terms are the same our original objective function for the stochastic model, with the first term concerning the costs for building and operating warehouses, the second term concerning transportation costs between customers and warehouses, and the third term concerning transportation costs between suppliers and warehouses.

The fourth term, however, is an extra term that penalizes not meeting all of our demand. When our total supply is less than our total demand (i.e. we don't meet all of our demand), then that final term is positive, increasing our objective function. This disincentivizes solutions that don't meet all of our demand, but because we no longer have the constraint (1), our solutions are no longer *required* to always meet demand in every scenario. This way, high-demand scenarios can find solutions that don't satisfy all of the demand (and thus are penalized for it), no longer forcing the formulation to over-build warehouses in order to accommodate for these few scenarios.

There is a trade-off between supplying more goods than necessary (and so lowering the objective function via our extra term) and having to pay for more transportation costs. If the penalty term is comparatively smaller than transportation costs, then the formulation would be incentivized to make each supplier reach maximum capacity. This wouldn't be sensible in a real world situation and so we would want to adjust λ in order to penalize at an appropriate amount. Equally, a λ variable that penalizes too harshly for not meeting all the demand will act too conservatively, falling victim to the same problem our original stochastic model did.